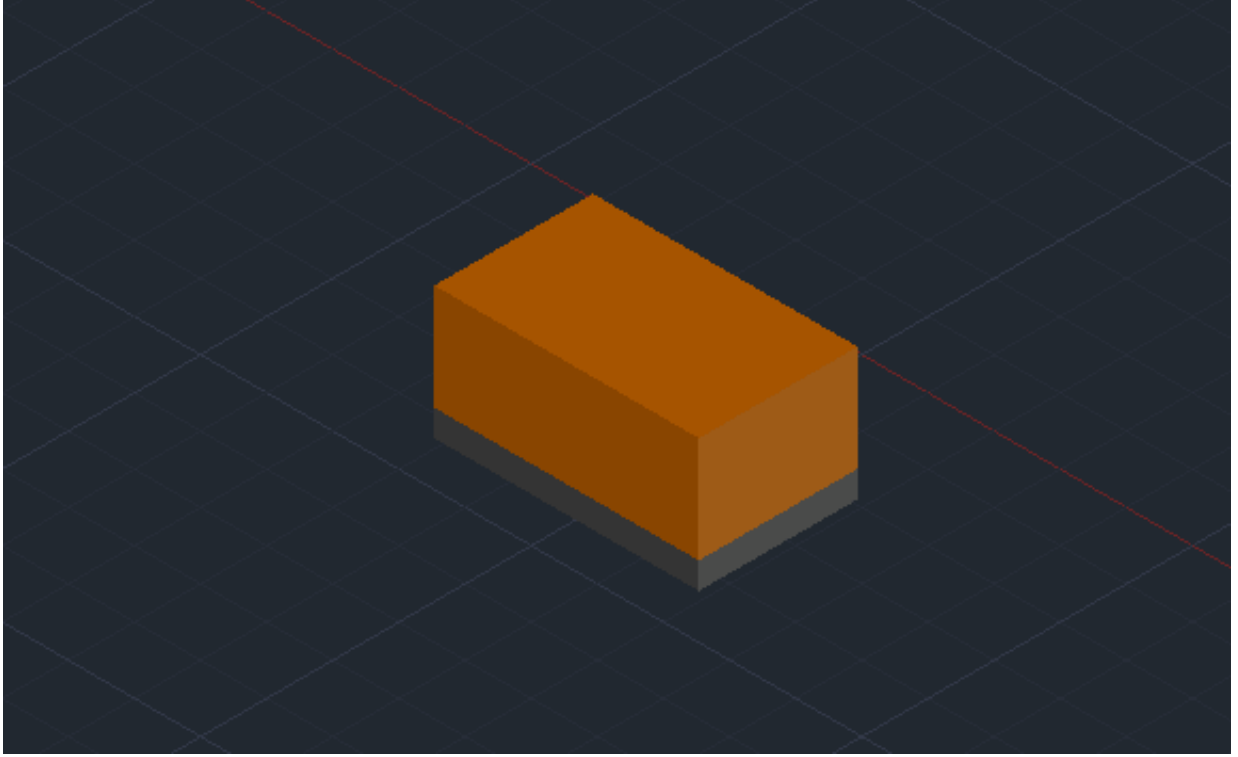


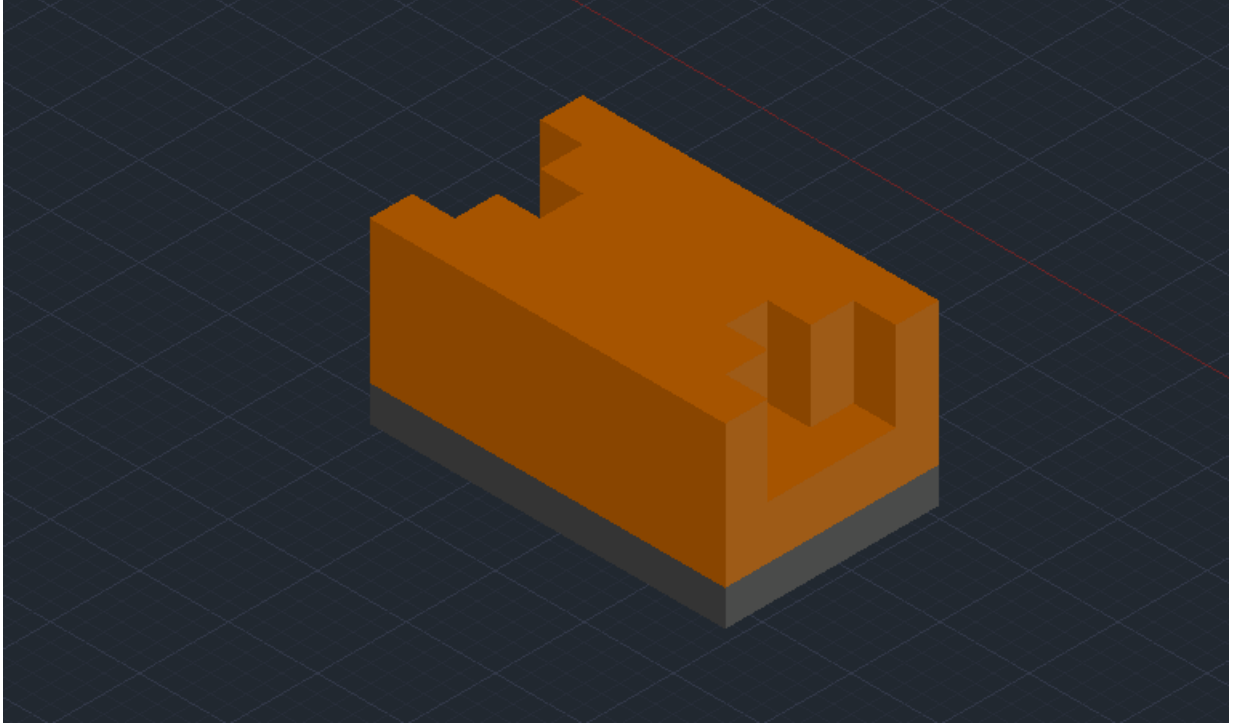
Semiconductor Homework #2

Preparer : Mustafa Yağcı

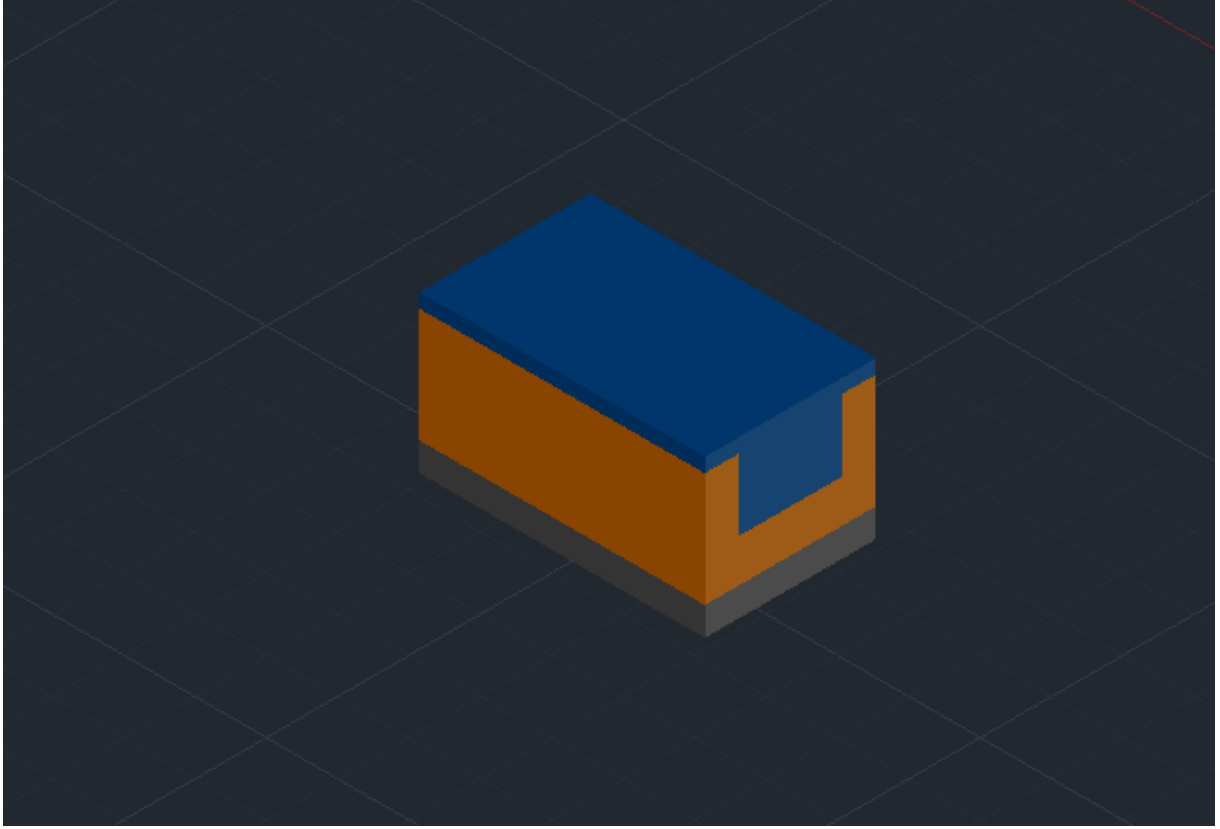
School No : 151220192027



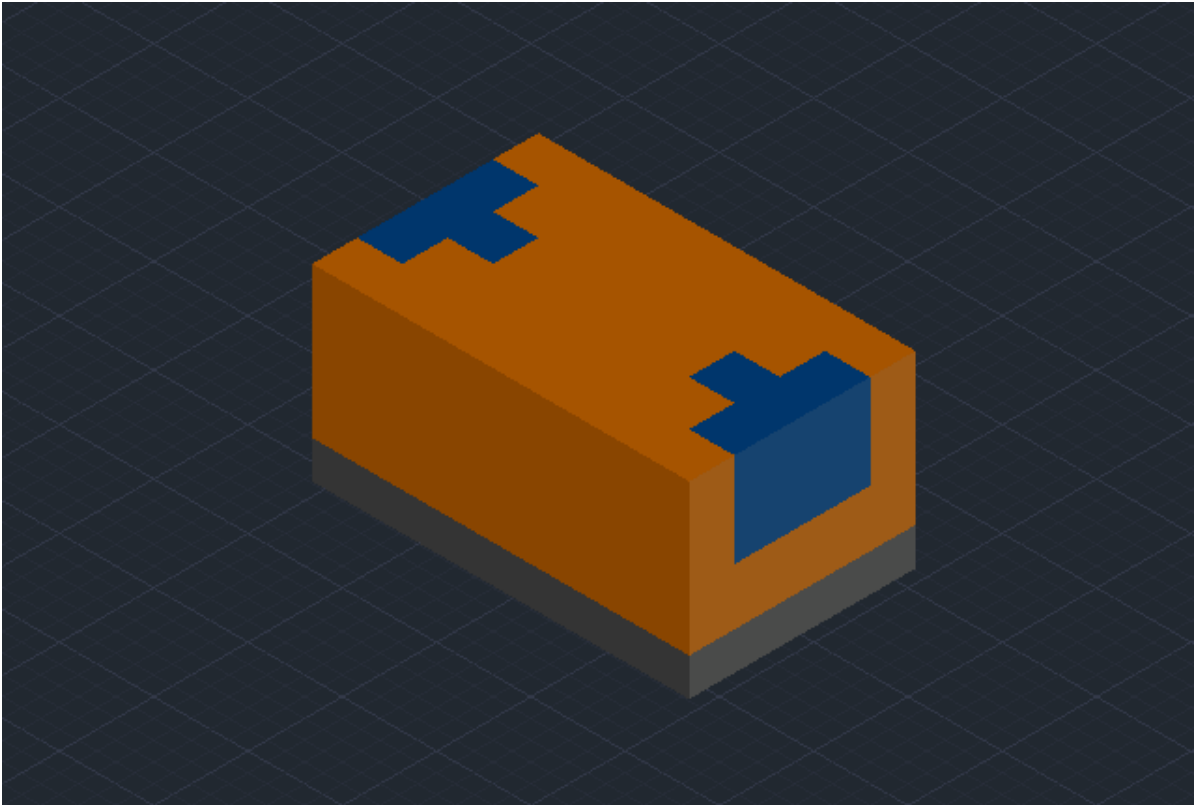
Şekil 1 : 700 nm Silicondioxide Growth (Orange Part)



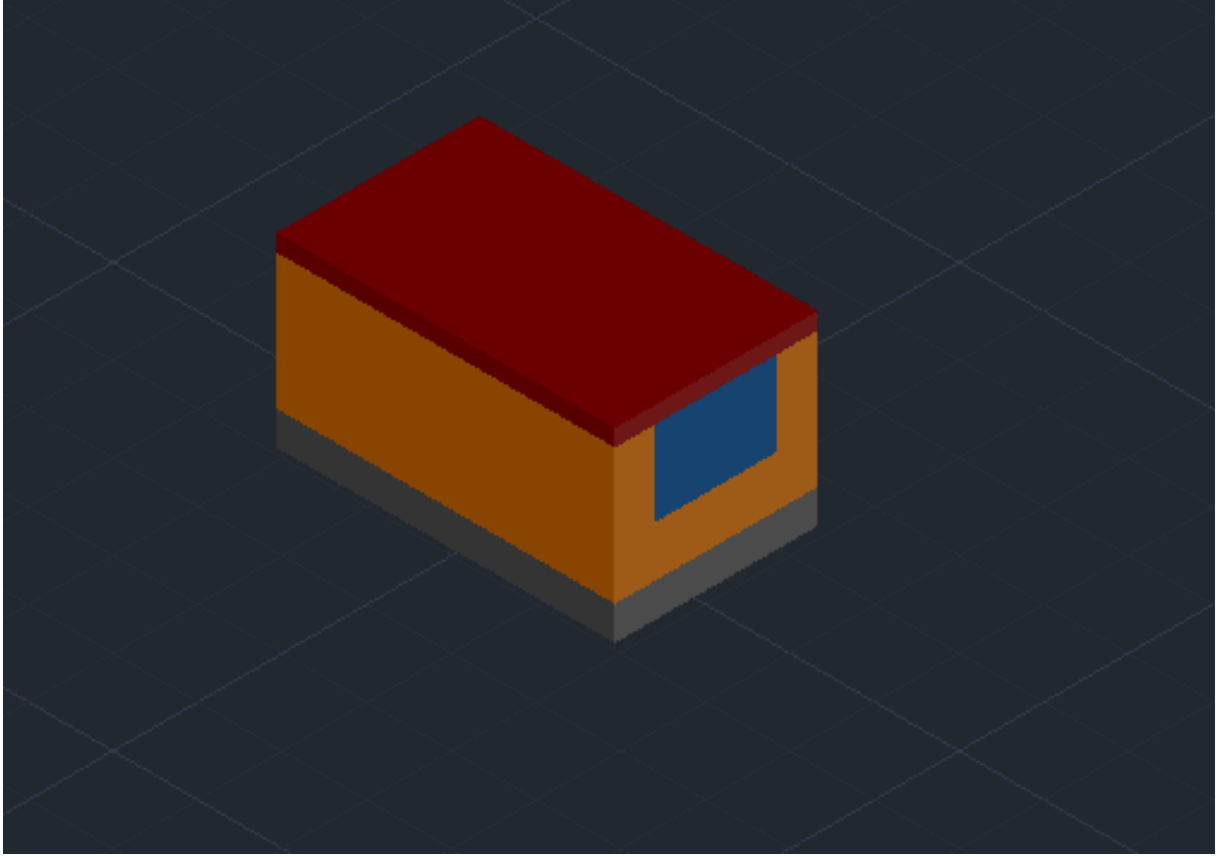
Şekil 2 : 250 nm Etch Silicondioxide (Deep Trench Formation)



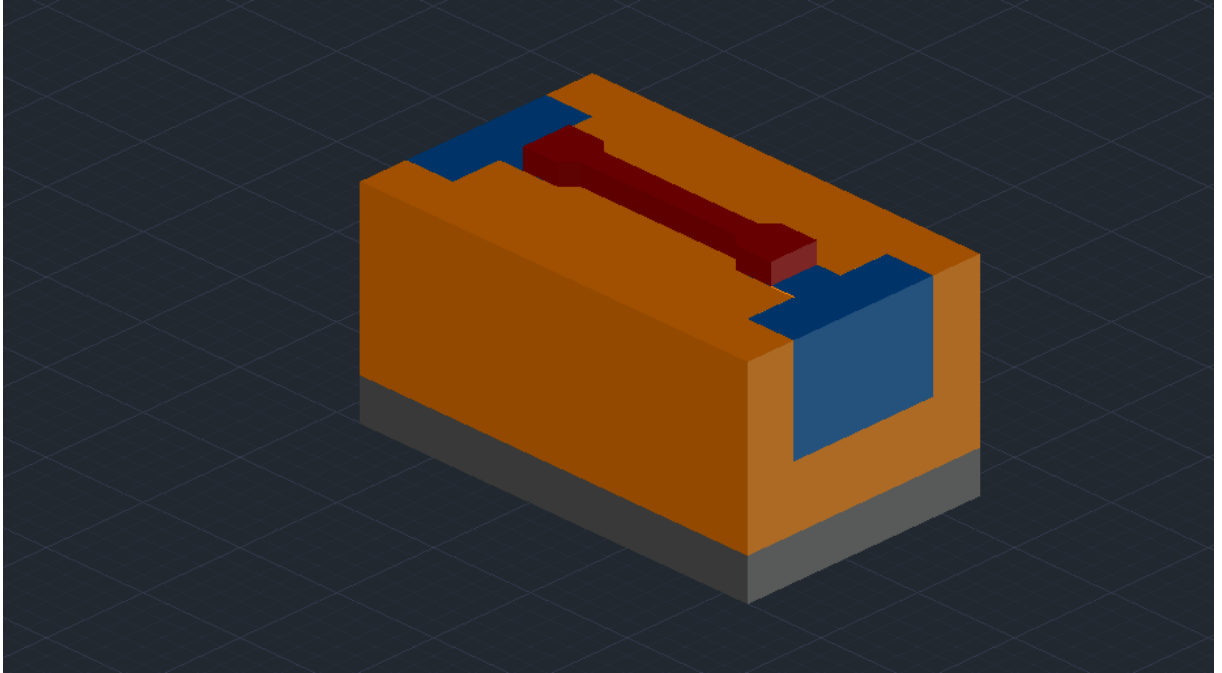
Şekil 3 : 300 nm dopped TIN (Blue Part)



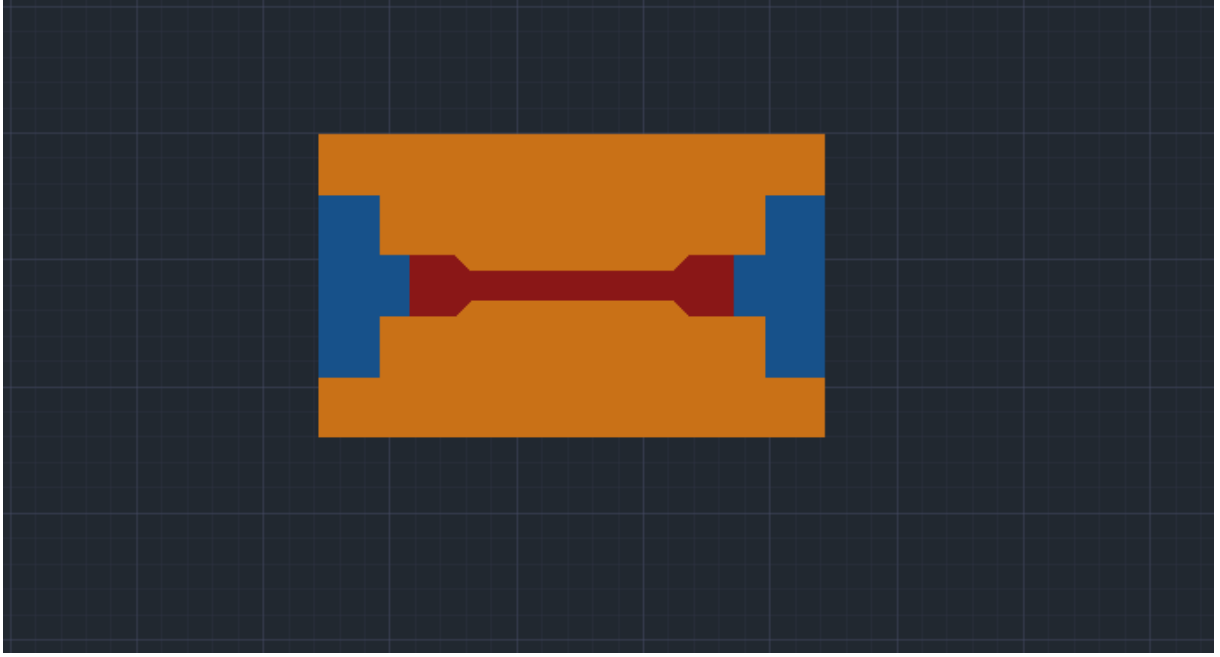
Şekil 4 : 50 nm TIN removed with CMP (Chemical Mechanical Planarization)



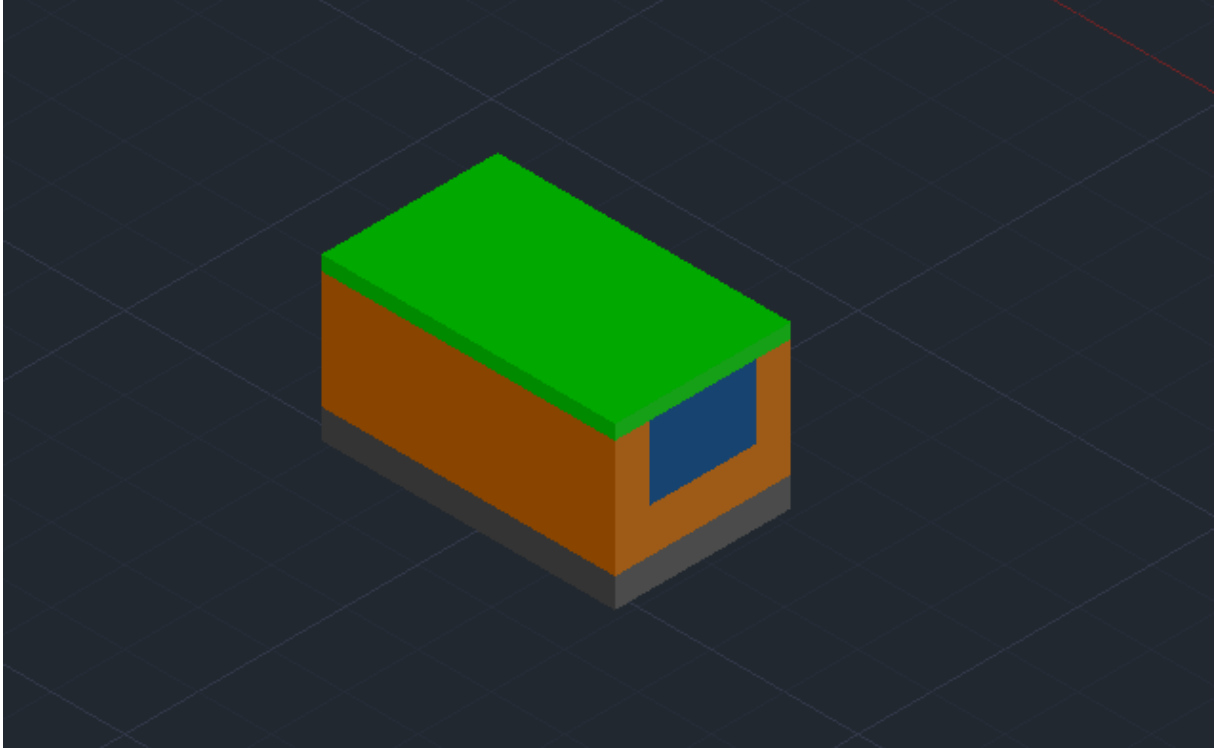
Şekil 5 : GST Film Deposition (Red Part)



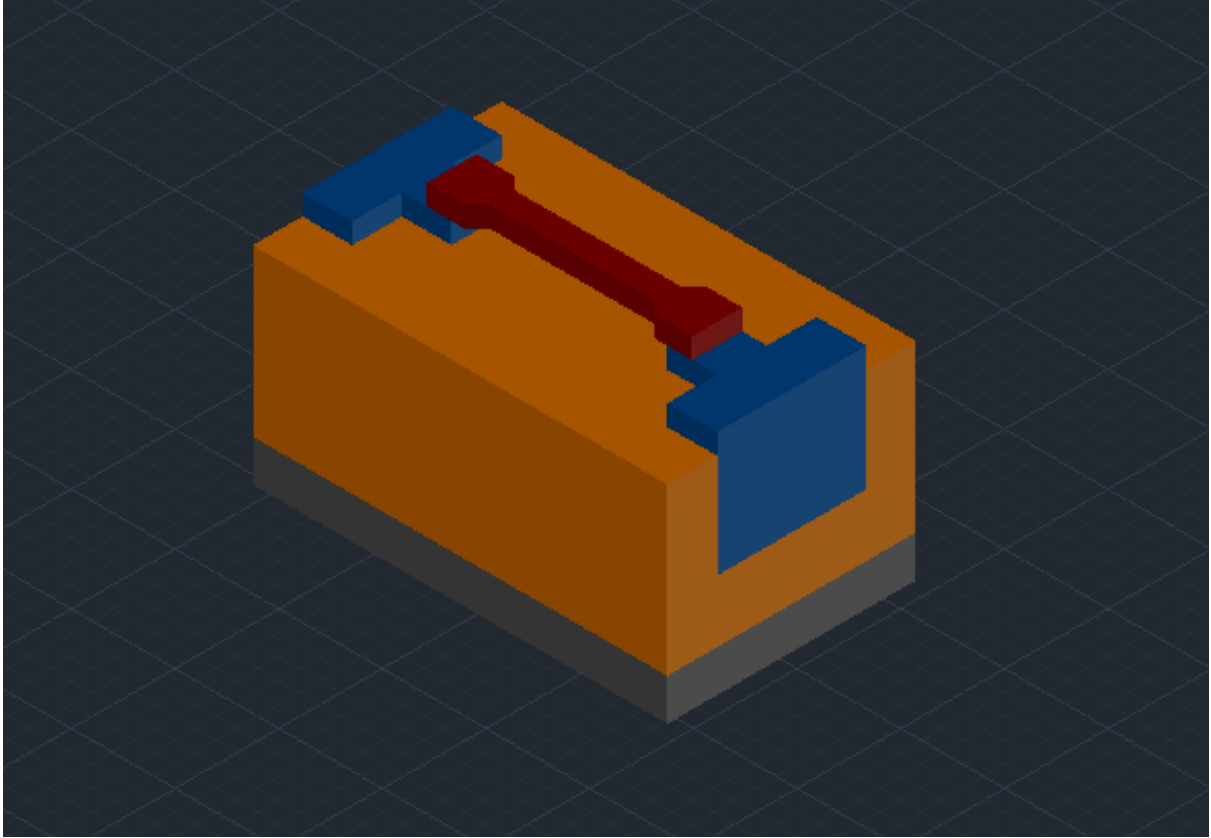
Şekil 6 : GST Film Patterning (Red Part)



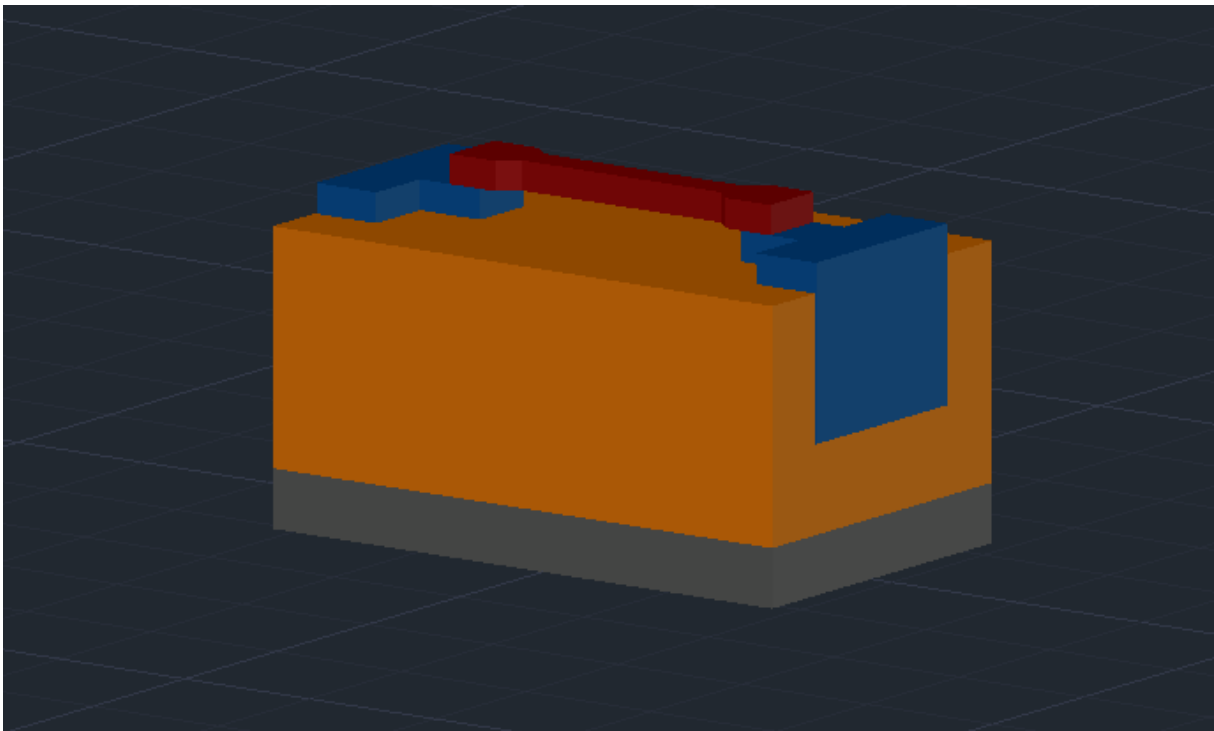
Şekil 7 : GST Film Patterning (Top View)



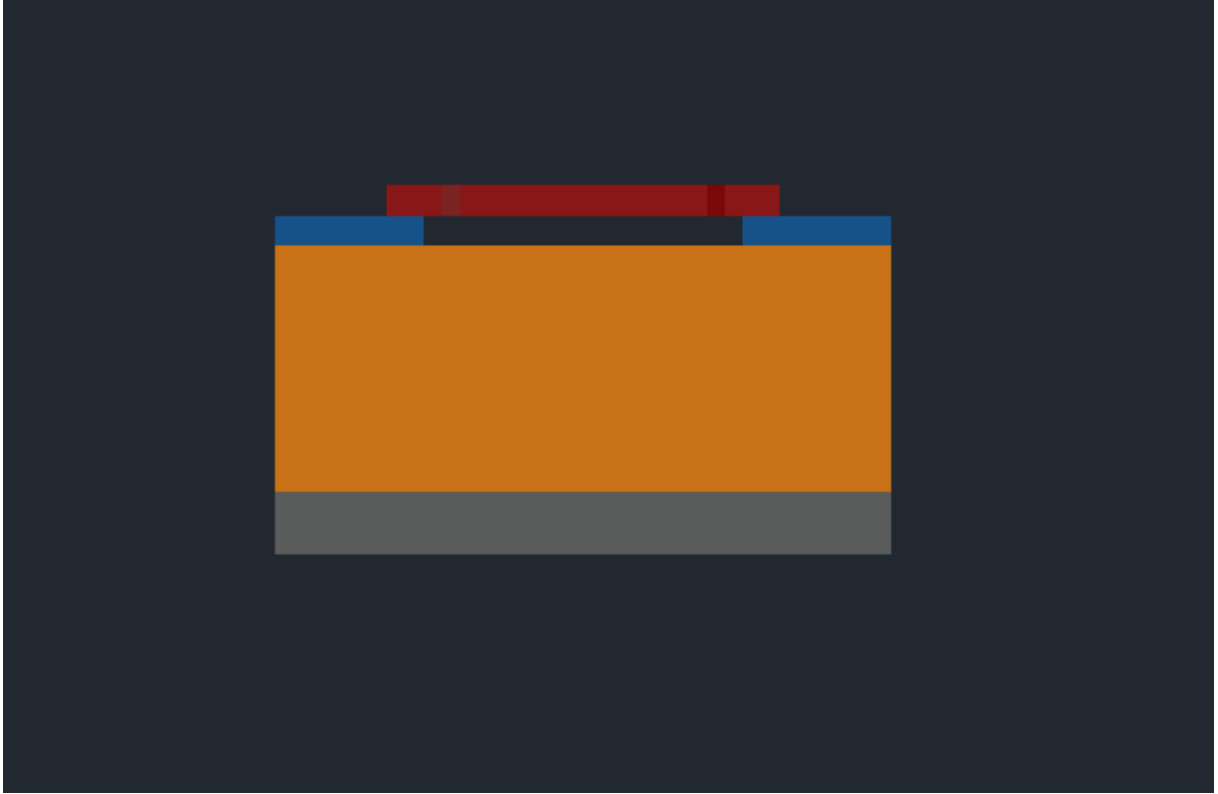
Şekil 8 : Si₂N₄ Cap Layer Deposition (Green Part)



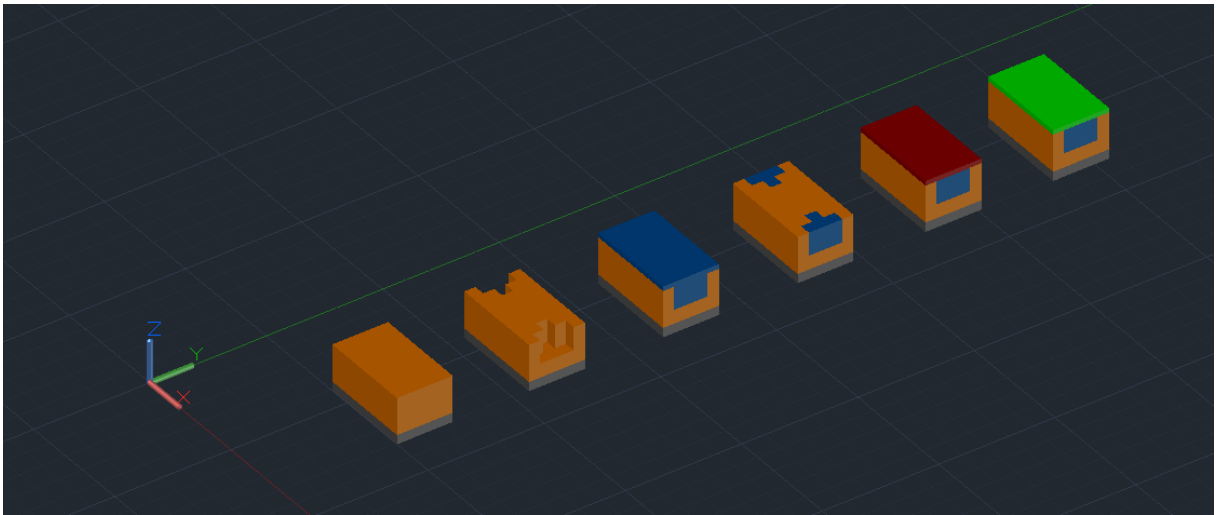
Şekil 9 : Pattern Detail Display (50 nm Silicondioxide Etched)



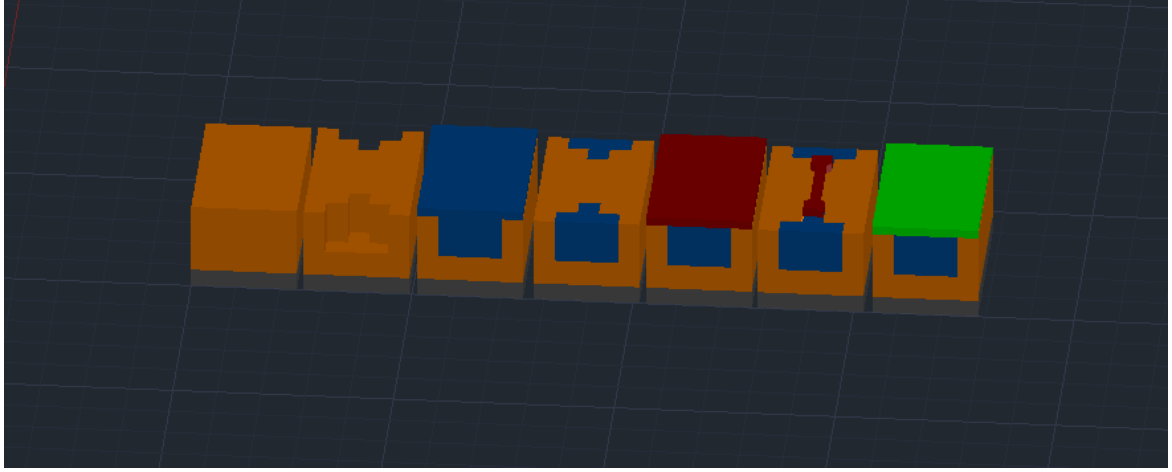
Şekil 10 : Another Pattern Detail Display



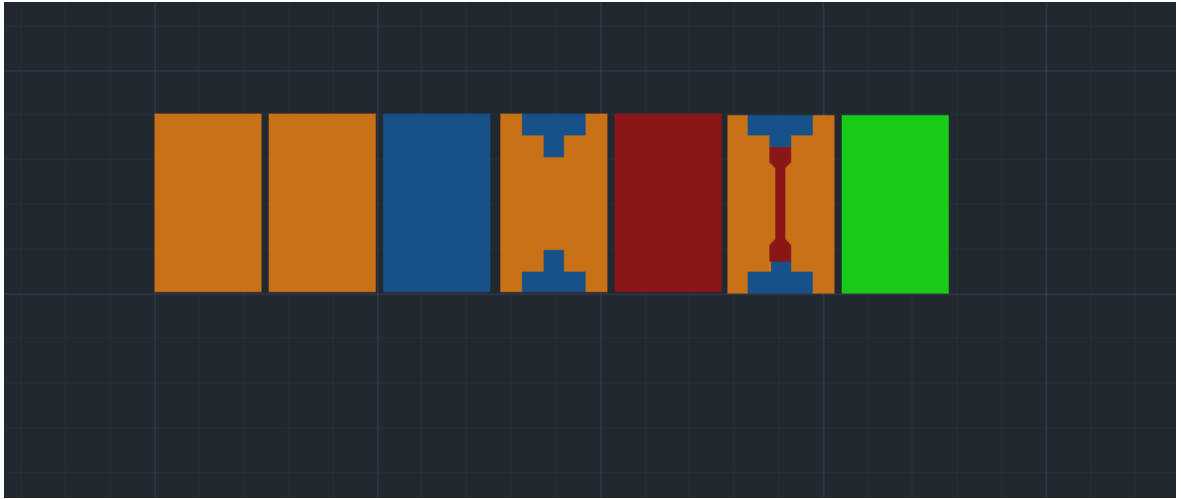
Şekil 11 : Side View Pattern Detail Display



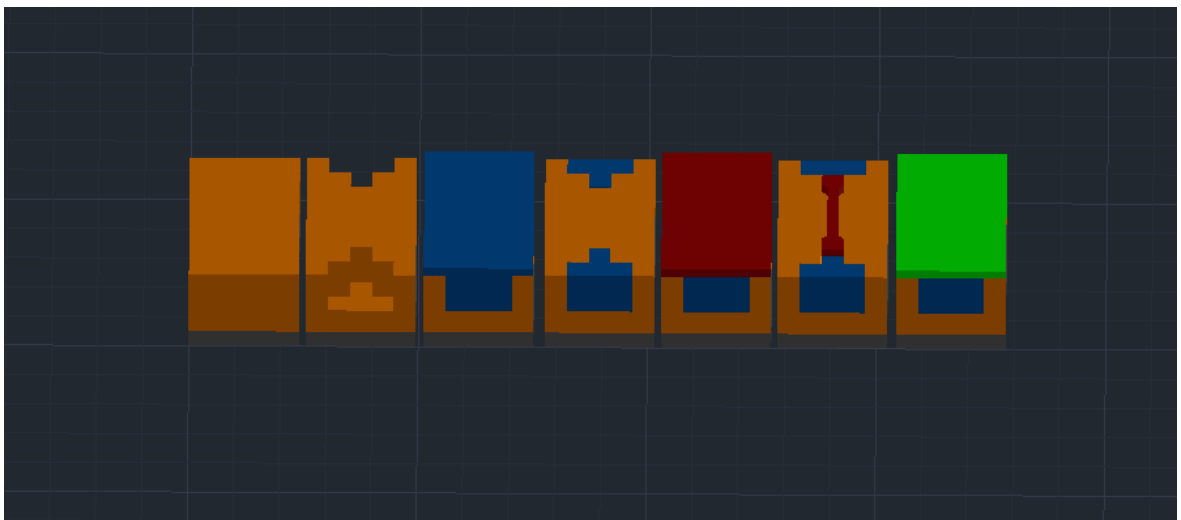
Şekil 12 : All GST Line Cell Fabrication Process



Şekil 13 : All GST Line Cell Fabrication Process



Şekil 14 : All GST Line Cell Fabrication Process (Top View)



Şekil 15 : All GST Line Cell Fabrication Process

References:

1. F. Dirisaglik, K. Cil, L. Adnane, M. Wennberg, A. King, A. Faraclas, M. B. Akbulut, Y. Zhu, C. Lam, A. Gokirmak, and H. Silva, "Electrical resistivity of liquid Ge₂Sb₂Te₅ based on thin-film and nanoscale device measurements," *IEEE Transactions on Electron Devices*, vol. 60, no. 1, pp. 433-440, Jan. 2013.
2. Cywar, F. Dirisaglik, M. Akbulut, G. Bakan, S. Steen, and H. Silva, "Scaling of silicon phase-change oscillators," *IEEE Electron Device Letters*, vol. 32, no. 11, pp. 1486–1488, Nov. 2011, doi: 10.1109/LED.2011.2164511.